

One exact bridge between two Leech coordinate models

The three marked Golay octads can be aligned with Wilson's three octonions by a signed permutation, not merely by an existence theorem.

For $\lambda = x/\sqrt{8}$, put $y_i = \epsilon_i x_{\pi(i)}$ and use the three blocks of $y/2$ as octonions.

	π							
0	0	1	3	4	8	23	7	16
1	10	22	15	18	12	21	19	20
2	2	13	5	11	6	14	17	9

$$\epsilon = (+ - - - - - - - \mid - - - - + + - + \mid - - - - + + - +).$$

The inverse is $x_{\pi(i)} = \epsilon_i y_i$. The map preserves the lattice and squared norm exactly. Thirty-seven generator tests plus unimodularity prove equality of the infinite lattices.

The cubic must be retained as well. Set $t = P\lambda$, $n_i = N(z_i)$ and $\nu = 2 \operatorname{Re}((z_0 z_1) z_2)$. In the full Albert state $\widehat{X} = \operatorname{diag}(t) + \iota(z)$,

$$\boxed{\begin{aligned} 4N_J(\widehat{X}) - pS_J(\widehat{X}) &= F_p(t) + \sum_i (p - 4t_i)n_i + 4\nu, \\ F_p(t) &= 4t_0 t_1 t_2 - p \sum_{i < j} t_i t_j. \end{aligned}}$$

A lattice isometry need not preserve this chosen cubic. Keeping the correction makes the interface exact rather than conflating the two invariants.

Full statement: An octad-aligned Golay–octonion coordinate bridge and its full cubic correction.